



Crypto Mispricing and Portfolio Diversification: Evidence from a Long–Short Strategy

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Abstract

This paper evaluates the role of an active crypto strategy in enhancing a traditional 60/40 equity–bond portfolio. We create a market-neutral long–short strategy built on a mispricing factor that uses both size and risk-adjusted momentum signals, computed from publicly available weekly data. The crypto universe is filtered for liquidity and availability, consisting of 79 assets from October 2019 to March 2025. The benchmark portfolio consists of SPY and AGG ETFs, and asset pricing factors conform to Fama–French and Carhart models.

The empirical results show that even modest but significant crypto exposure via this systematic strategy leads to a positive marginal contribution to portfolio performance by increasing Sharpe and Sortino ratios, and by reducing drawdowns. Factor regressions reveal statistically and economically significant alpha, not attributable to conventional risk premia. We verified robustness through out-of-sample tests, bootstraps, and stress scenarios, confirming the effective diversification properties of a transparent and replicable crypto strategy.

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1 Introduction

Following recent geopolitical and economic shifts, including rising skepticism toward traditional financial institutions in the post-Trump era, cryptocurrencies have gained unprecedented attention as both speculative assets and potential long-term stores of value. This evolution has been echoed even at the highest levels of institutional finance: in his 2025 annual letter to investors, BlackRock CEO Larry Fink warned that if the United States fails to control its fiscal deficit and rising national debt, the U.S. dollar could risk losing its global reserve currency status to digital assets such as Bitcoin.

This provocative statement captures a broader sentiment: digital assets, once viewed as fringe instruments, are now forcing a rethinking of strategic asset allocation. Their appeal lies in their decentralization, narrative-driven dynamics, and low correlation with traditional financial markets. Yet these same features raise questions about their integration into established portfolio frameworks such as the 60/40 model, a benchmark allocation based on 60% equities and 40% bonds that has long represented the foundation of modern portfolio theory. The central question we pose is whether it is meaningful, both economically and statistically, to incorporate an active crypto strategy into such a portfolio.

To address this, we construct and analyze a systematic, long-short cryptocurrency strategy based on a mispricing factor that captures deviations between an asset’s observed price and a theoretically implied valuation proxy. In the absence of universally accepted fundamental anchors, we operationalize mispricing by combining two intuitive cross-sectional signals: size (defined as the inverse logarithm of market capitalization) and risk-adjusted momentum (RA-MOM), calculated as cumulative return divided by recent return volatility, measured over 1-, 2-, and 4-week horizons. These signals are standardized via z-scores and averaged into a composite mispricing score, which is used to rank a broad crypto universe on a weekly basis.

Each week, we construct a market-neutral portfolio that goes long on the 30% of assets with the highest mispricing scores (i.e., the most “undervalued”) and short on the 30% with the lowest scores (i.e., the most “overvalued”). This approach draws inspiration from the empirical finance literature on style factors, most notably Fama and French (1993)

and Carhart (1997), while adapting to the behavioral inefficiencies and speculative nature of the crypto market.

Building on this literature, recent findings by Han et al. (2024) further reinforce the relevance of factor-based investing in the cryptocurrency space. Their study shows that portfolios based on signals such as size, momentum, and illiquidity signals stochastically dominate naive strategies, offering significant diversification benefits. Unlike many models that rely on high-frequency or proprietary data, our approach is transparent, low-frequency, and replicable.

To assess the practical relevance of our strategy, we embed it within a traditional 60/40 portfolio composed of SPY (equities) and AGG (bonds), rebalanced monthly. We explore a range of crypto allocations from 1% to 20%, funded by proportionally reducing both equity and bond weights. This structure allows us to test whether, and to what extent, even modest crypto exposure improves overall portfolio performance and risk characteristics.

From a methodological perspective, we complement performance evaluation with a rigorous econometric framework. We estimate multi-factor regressions based on the Fama–French three-factor model augmented with momentum to assess the presence of statistically significant alpha and control for systematic risk exposures. Diagnostic tests include the Chow test (structural breaks), Breusch–Pagan (heteroskedasticity), and Durbin–Watson and Ljung–Box (autocorrelation).

To validate our findings, we conduct out-of-sample testing using a training-test split. We apply bootstrap resampling to estimate confidence intervals and simulate stress scenarios, such as a 30% weekly drawdown in crypto markets, to evaluate portfolio robustness under extreme conditions.

Our contribution is twofold. Empirically, we demonstrate that a transparent, composite mispricing factor, built from standardized size and risk-adjusted momentum signals, can generate economically meaningful alpha within the cryptocurrency market. Methodologically, we integrate this signal into a traditional 60/40 portfolio and evaluate its performance using a rigorous econometric framework that includes multi-factor regressions, diagnostic tests, out-of-sample validation, and stress testing. This provides a replicable blueprint for incorporating speculative assets into conventional investment strategies without relying on proprietary models or high-frequency data.

We structure the remainder of the paper as follows: Section 2 surveys the relevant literature on portfolio theory, cryptocurrency markets, and cross-sectional factor models; Section 3 describes the data sources and preprocessing steps; Section 4 outlines the construction of the benchmark portfolio and the mispricing-based crypto strategy; Section 5 details the econometric methodology and robustness diagnostics; Section 6 presents and discusses the empirical results; Section 7 concludes with key insights and future research directions.

2 Context and Foundations of the Strategy

2.1 The 60/40 Portfolio as a Benchmark

We begin by constructing a traditional 60/40 portfolio, long considered a cornerstone of modern portfolio theory. This allocation, composed of 60% equities and 40% bonds, is widely adopted by both institutional and retail investors for its balance between growth and capital preservation. In our implementation, the equity exposure is captured through the SPDR S&P 500 ETF Trust (SPY), while fixed income is represented by the iShares Core U.S. Aggregate Bond ETF (AGG), both of which serve as liquid proxies for their respective asset classes.

Using historical daily prices, we simulate the portfolio’s value starting from a normalized capital of 1, computing log returns and applying monthly rebalancing to maintain target weights and prevent allocation drift over time.

Standard risk-return metrics, such as annualized return, volatility, Sharpe ratio, maximum drawdown, and rolling statistics, are calculated solely to characterize the benchmark’s stability and performance under normal conditions.

Importantly, this 60/40 portfolio is not the focus of our analysis. It serves as a neutral baseline to systematically evaluate the incremental impact of integrating a disciplined, factor-based cryptocurrency overlay aimed at enhancing portfolio efficiency without materially increasing downside risk.

2.2 Theoretical Rationale and Mispricing Factor Construction

Cryptocurrencies have emerged as a distinct and increasingly influential asset class. Despite their extreme volatility and speculative nature, they exhibit low correlation with traditional markets, making them potentially valuable for diversification. However, their lack of fundamental anchors, illiquidity in many cases, and strong behavioral drivers pose unique challenges for portfolio integration.

Multiple studies suggest that cross-sectional anomalies observed in equity markets may also be present in cryptoassets. Notably, Liu, Tsyvinski, and Wu (2019) document persistent return predictability based on momentum and reversal effects, even in the absence of conventional valuation anchors. Complementary evidence is provided by Borri (2022), who demonstrates the presence of systematic factor risk in cryptocurrency returns. Building on these findings, Borri et al. (2024) highlight that a single cross-sectional factor can explain a substantial portion of return variations across cryptoassets, further reinforcing the relevance of structured factor-based approaches in cryptoasset pricing. Together, these insights open the door to systematic strategies that exploit persistent mispricing patterns.

Our approach builds on the framework developed by Han et al. (2021), who propose a composite mispricing factor for cryptocurrencies. We extend their work by embedding the strategy within a broader portfolio context and subjecting it to rigorous econometric validation.

The factor construction relies on two standardized signals:

- **Size**, defined as the inverse of the logarithm of market capitalization. This transformation emphasizes smaller-cap assets, which tend to be more mispriced due to lower liquidity, weaker analyst coverage, and greater retail participation.
- **Risk-Adjusted Momentum (RA-MOM)**, computed over 1-, 2-, and 4-week horizons. For each asset, cumulative returns over each period are divided by realized volatility to penalize noisy price swings and highlight stable upward trends.

Each signal is transformed into a z-score, and the average of the size and the three momentum components yields the final mispricing score. On a weekly basis, we rank the

universe of cryptoassets based on this score.

Each week, we construct a market-neutral long–short portfolio that goes long on the top 30% and short on the bottom 30% of assets, based on their mispricing scores. The selection is updated weekly and applied only to cryptocurrencies with valid signal and return data at each rebalancing date.

Unlike many proprietary or high-frequency approaches, our methodology is simple, transparent, and replicable using publicly available data. This allows us to evaluate whether style-based strategies from equity markets can be successfully transferred to a digital asset context.

Our work also contributes to a broader literature on crypto asset pricing. In addition to Liu et al. (2019), Blau et al. (2021) explore long–short strategies based on liquidity and technical indicators, reporting persistent abnormal returns at weekly frequencies. However, many such models suffer from overfitting, lack transparency, or are difficult to implement in practice.

By contrast, our framework prioritizes interpretability and robustness. We analyze both in-sample and out-of-sample performance, assess regime sensitivity, and consider real-world implementation constraints. Importantly, we test whether the mispricing strategy can be successfully embedded within a 60/40 portfolio while preserving desirable risk–return characteristics.

To evaluate this integration, we conduct time series regressions of portfolio excess returns on the Fama–French three-factor model plus momentum. We estimate alpha and beta coefficients, test for structural breaks (Chow), heteroskedasticity (Breusch–Pagan), and autocorrelation (Durbin–Watson and Ljung–Box), and compute rolling Sharpe, Sortino, and volatility metrics.

Robustness is further ensured through out-of-sample validation, bootstrap resampling for inference, and simulation of extreme downside scenarios (e.g., a sudden 30% weekly drop in crypto markets). We also implement the Diebold–Mariano test to compare predictive accuracy across alternative allocations.

Taken together, these tools allow us to rigorously assess whether cryptocurrencies, selected via mispricing filters, can offer uncorrelated alpha within a traditional multi-asset

framework.

3 Data Description

The empirical analysis is based on three main data sources:

- Historical cryptocurrency data collected via API;
- Daily adjusted prices of benchmark ETFs representing the equity and bond markets;
- Asset pricing factors from the Fama–French and Carhart momentum models.

The sample period covers January 1, 2018, to March 28, 2025, for traditional asset classes (SPY and AGG) and factor models (Fama–French and Momentum), and extends from October 23, 2019, to March 29, 2025, for cryptocurrency data and digital asset strategies. This temporal structure captures multiple episodes of heightened volatility and macro-financial regime shifts, ensuring robust econometric inference and comprehensive performance evaluation.

3.1 Cryptocurrency Data

Cryptocurrency data were sourced from CryptoCompare, a reputable and widely used aggregator of digital asset market information. The dataset spans January 1, 2019, to March 29, 2025, with daily frequency. Using the CryptoCompare API and a verified developer key, we retrieved historical closing prices and market capitalizations for a curated universe of 79 cryptocurrencies, selected based on liquidity, availability, and historical completeness.

To ensure data quality and consistency, we applied the following filtering criteria:

- Exclusion of stablecoins, utility tokens, and assets with missing or inconsistent records;
- Retention of assets with uninterrupted time series across the entire sample period;
- Removal of NaN or Inf values, ensuring synchronized timestamps and consistent date formats.

Each asset’s daily log-returns were computed from closing prices and then aggregated to weekly frequency, matching the rebalancing horizon of the trading strategy. Concurrently, we computed market capitalization-based size proxies and risk-adjusted momentum (RA-MOM) signals, which jointly form the composite mispricing score (detailed in Section 2.2). This dataset provides a rich cross-section of cryptoassets over multiple years, allowing for both cross-sectional and time-series inference under varying market conditions.

Selection Bias Notice. It is important to acknowledge a potential selection bias arising from the way the cryptocurrency universe was constructed. Our filtering process is based on the set of cryptocurrencies that are currently active and meet data availability and liquidity criteria *as of 2025*. This ex-post approach implies that cryptoassets which were relevant in the early part of the sample period but later became illiquid, delisted, or failed are not included in our dataset. As a result, the analysis may be subject to a survivorship bias, which could artificially inflate the perceived robustness and profitability of the strategy. While our objective was to evaluate implementable strategies on currently investable assets, we explicitly recognize this limitation in the data construction process.

3.2 Traditional Benchmark: SPY and AGG ETFs

To construct the traditional benchmark portfolio, we use two highly liquid and widely adopted ETFs:

- **SPDR S&P 500 ETF Trust (SPY):** tracks the performance of the S&P 500 index, capturing U.S. large-cap equity exposure.
- **iShares Core U.S. Aggregate Bond ETF (AGG):** tracks the Bloomberg U.S. Aggregate Bond Index, representing investment-grade U.S. fixed income.

We retrieved daily adjusted closing prices for both ETFs over the period from January 1, 2018, to March 28, 2025. The prices were converted into log-returns, and all dates were aligned to ensure consistency across the dataset.

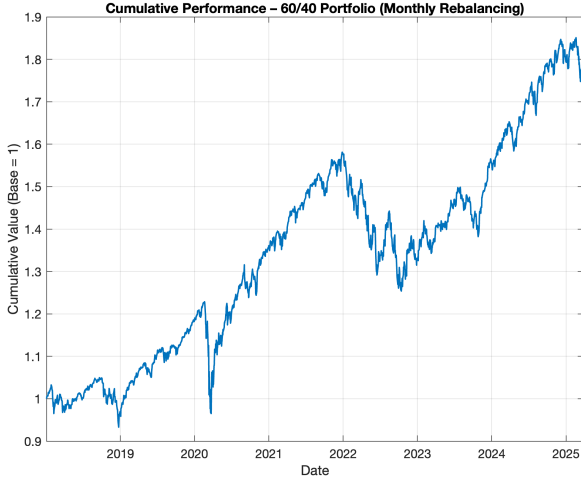
The benchmark portfolio is initialized with a value of 1 and rebalanced monthly to maintain the target 60% equity (SPY) and 40% bond (AGG) allocation. This procedure prevents portfolio drift and ensures stable asset exposure over time.

The benchmark portfolio exhibits moderate long-term growth and reduced volatility compared to pure equity exposure. The annualized Sharpe Ratio of 0.650 indicates strong risk-adjusted performance relative to conventional efficiency thresholds (e.g., 0.5).

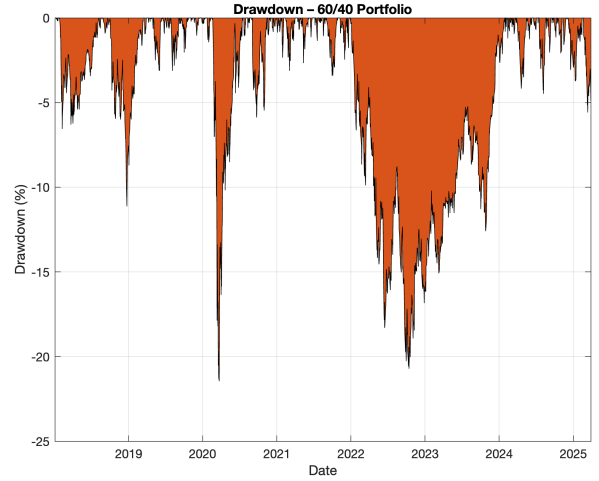
Visual diagnostics reinforce these insights and provide an overview of the 60/40 benchmark behavior:

- **Figure 1a** displays the cumulative returns of the 60/40 portfolio, rebalanced monthly, over the sample period.
- **Figure 1b** illustrates the portfolio's historical drawdowns, capturing periods of maximum decline from peak value.
- **Figure 1c** shows the 1-year rolling Sharpe Ratio, reflecting the stability of risk-adjusted performance over time.
- **Figure 1d** presents a boxplot of monthly log-returns for the 60/40 portfolio, including the 5% Value-at-Risk (VaR) threshold, to visualize downside risk and distribution asymmetry.

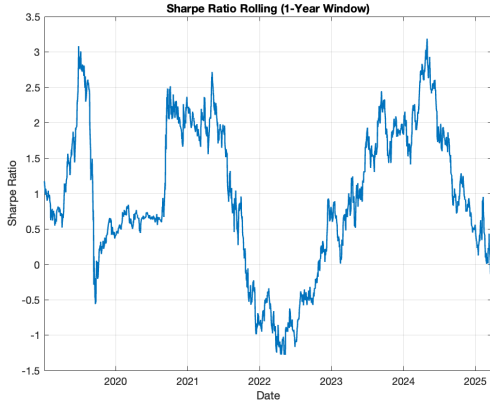
All figures are based on weekly or monthly log-returns and standardized to ensure consistency. The 60/40 portfolio serves as the baseline allocation throughout the study, against which crypto-enhanced strategies are evaluated.



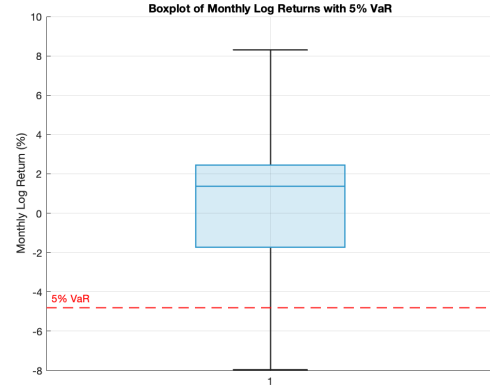
(a) Cumulative returns of the 60/40 portfolio (monthly rebalancing)



(b) Drawdown history of the 60/40 portfolio



(c) 1-year rolling Sharpe Ratio for the 60/40 portfolio



(d) Monthly log-returns with 5% VaR threshold (60/40 portfolio)

Figure 1: Visual diagnostics for the 60/40 benchmark portfolio.

This benchmark portfolio serves as the reference allocation for evaluating the integration of a systematic crypto component. It is representative of the standard passive strategy underlying many target-date and risk-balanced investment vehicles.

3.3 Fama–French Factors and Data Preparation

To account for systematic risk exposures and isolate potential alpha, we incorporate four well-established risk factors in our empirical analysis: the Fama–French three-factor model (Fama and French, 1993) and the Carhart momentum factor (Carhart, 1997).

Specifically, we use the following daily time series, retrieved from the Kenneth R. French data library:

- **Mkt-RF:** Market excess return
- **SMB:** Size factor (Small minus Big)
- **HML:** Value factor (High minus Low book-to-market)
- **MOM:** Momentum factor

Each factor is provided in daily frequency, covering the period from January 2018 to December 2024, and is expressed in percentage returns. We convert the factors into weekly frequency by summing the daily returns within each calendar week. This ensures temporal consistency with the weekly return series used for both the crypto strategy and the benchmark portfolio, enabling reliable comparative analysis.

The risk-free rate (**RF**) is also obtained from the same source and used to compute weekly excess returns for all portfolios and strategies throughout the analysis.

3.4 Data Cleaning and Synchronization

Prior to model estimation and strategy implementation, we applied a series of data cleaning and synchronization steps to ensure consistency across sources:

- **Stablecoin exclusion:** The initial cryptocurrency symbol list was manually filtered to exclude all stablecoins and pseudo-stable assets (e.g., USDT, USDC, DAI), which do not exhibit meaningful price dynamics.
- **Token validation:** Only cryptoassets with complete price and market capitalization data from October 2019 to March 2025 were retained. Assets with excessive missing data, low liquidity, or irregular trading patterns were removed.
- **Frequency harmonization:** All financial time series, including ETF prices, crypto prices, and factor returns, were converted to weekly frequency, consistent with the strategy’s rebalancing interval.
- **Return and signal computation:** Weekly log returns were computed. Signals based on size and risk-adjusted momentum (RA-MOM) were standardized cross-sectionally using z-scores to ensure comparability across assets and dates.

- **Score construction:** The final mispricing score was defined as the average of the standardized size and RA-MOM signals. This score was used to rank cryptoassets each week (see Section 2.2).
- **Date alignment:** All datasets were merged using the intersection of weekly timestamps, ensuring that only aligned and complete observations entered the regression and portfolio construction phases.

These rigorous data handling procedures support reproducibility and help prevent issues such as look-ahead bias, interpolation artifacts, or inconsistencies in sample coverage.

3.5 Interpretation of Benchmark Performance and the Risk–Return Trade-Off

The performance statistics confirm that the 60/40 portfolio offers a balanced trade-off between return and risk. With an annualized return of 7.84% and volatility of 12.06%, the portfolio achieves a Sharpe ratio of 0.650, surpassing the commonly accepted threshold of 0.5 that denotes favorable risk-adjusted returns.

The maximum drawdown of –21.46% and the monthly 5% Value at Risk (VaR) of –4.68% highlight significant yet manageable downside risks. These risks remain within expected bounds given the turbulent sample period (2018–2025), characterized by unprecedented events such as the COVID-19 pandemic and the geopolitical shock from the Russia–Ukraine conflict.

When compared to two benchmark extremes—a fully equity-based portfolio (100% SPY) and a fully bond-based portfolio (100% AGG)—the 60/40 allocation clearly demonstrates superior risk efficiency:

- **SPY (100% Equity):** Highest return (11.67%) but accompanied by the highest volatility (19.44%) and maximum drawdown (–33.72%), resulting in a lower Sharpe ratio (0.600).
- **AGG (100% Bonds):** Lowest volatility (5.96%) and lowest return (1.30%), with a Sharpe ratio of only 0.218, insufficient given the low yield.

- **60/40 Portfolio:** Strikes a balanced middle ground, significantly reducing downside risk while maintaining attractive returns. Importantly, it has the highest Sharpe ratio among the three portfolios, indicating optimal risk-adjusted performance.

The VaR comparison reinforces this view: the 60/40 portfolio’s monthly VaR of -4.68% comfortably sits between SPY (-8.35%) and AGG (-2.69%), reflecting its balanced risk profile.

Figure 1a (Cumulative Returns) further illustrates these observations. While SPY delivers the highest cumulative returns, its trajectory is punctuated by pronounced volatility and sharp drawdowns. Conversely, AGG offers stability but limited growth potential. The 60/40 portfolio achieves a smoother return profile characterized by moderate drawdowns and quicker recoveries.

These empirical findings underscore the balanced nature of the 60/40 portfolio, demonstrating substantial equity participation alongside meaningful downside protection. The favorable risk metrics and superior risk-adjusted returns establish this allocation as a robust benchmark. Against this backdrop, the following section introduces our systematic crypto strategy based on mispricing signals and details the empirical and econometric framework employed to assess its incremental performance contribution.

4 Methodology

4.1 Reference Benchmark: The 60/40 Portfolio

The benchmark used throughout this study is a traditional 60/40 portfolio, consisting of 60% equities and 40% investment-grade bonds. This allocation, long considered a foundational standard in modern portfolio theory, serves as a neutral baseline to evaluate the incremental value introduced by our crypto-based strategy.

As previously described, the equity and bond exposures are captured respectively by SPY and AGG, widely used proxies for their asset classes.

We collected daily adjusted closing prices for both ETFs over the period from January

1, 2018 to March 28, 2025, and computed daily log returns. To ensure consistency with the weekly rebalancing frequency of the crypto strategy, all returns were aggregated to the weekly level.

The portfolio is rebalanced monthly to preserve the target weights, effectively controlling for drift and maintaining the intended risk profile throughout the analysis period.

We do not explicitly apply transaction costs to the 60/40 portfolio for two primary reasons. First, SPY and AGG are highly liquid, low-turnover ETFs characterized by minimal bid-ask spreads and negligible management fees, thus implying minimal real-world frictions for passive investors. Second, our primary aim is to utilize the 60/40 allocation as a clean, frictionless benchmark, allowing us to properly isolate and assess the impact of the active crypto overlay, which involves higher turnover and consequently realistic rebalancing costs (as discussed in Section 5.2).

Over the entire sample period, the 60/40 portfolio delivered the following performance metrics:

Table 1: Performance summary of the 60/40 benchmark portfolio (2018–2025)

Metric	Value
Annualized Return	7.84%
Annualized Volatility	12.06%
Sharpe Ratio	0.650
Maximum Drawdown	−21.46%
5% Monthly VaR	−4.68%

These metrics confirm the portfolio’s intended profile: a moderate risk-return trade-off characterized by lower volatility than pure equities and higher return efficiency relative to bonds alone. Its Sharpe ratio of 0.650 notably surpasses the conventional threshold (0.5), indicating favorable risk-adjusted performance.

While informative, these metrics are intended solely to characterize the passive benchmark. The focus of this study lies in evaluating how the integration of a disciplined crypto-based overlay modifies portfolio performance, risk exposures, and diversification properties relative to this baseline.

Throughout this study, the 60/40 portfolio is employed as a benchmark in three distinct capacities:

- As a standalone passive allocation, representing a commonly utilized default strategy within institutional portfolios.
- As the foundational structure into which the crypto strategy is integrated, in varying proportions (1% to 20%), to assess potential diversification benefits and performance enhancements.
- As a control portfolio within statistical analyses designed to test alpha generation, systematic risk exposures, and robustness.

This structured benchmarking approach enables a clear evaluation of the mispricing-based crypto strategy’s impact across multiple dimensions, including return, volatility, and statistical significance.

4.2 Mispricing Strategy: Signal Construction and Weekly Portfolio Implementation

We implement a systematic long-short cryptocurrency strategy that exploits persistent cross-sectional mispricing patterns. The strategy is market-neutral and rebalanced weekly, based on a composite signal derived from two economically grounded and empirically validated indicators: *size* and *risk-adjusted momentum* (RA-MOM). Both signals are computed from daily data and standardized weekly to align with the rebalancing frequency of the strategy.

Size Signal

The size signal captures the inverse relationship between market capitalization and expected return, consistent with the small-minus-big (SMB) anomaly observed in equity markets. For each asset i and week t , the signal is defined as:

$$\text{Size}_{i,t} = -\log(\text{MC}_{i,t} + \varepsilon) \quad (1)$$

where $\text{MC}_{i,t}$ is the weekly average of daily market capitalization (proxied via “volumeto”), and ε is a small constant to ensure numerical stability (e.g., 10^{-8}). The signal is computed in MATLAB using `retime(..., 'weekly', @mean)`.

Risk-Adjusted Momentum (RA-MOM)

To capture short-term momentum effects while penalizing volatility, we compute RA-MOM over three rolling horizons: 1 week (7 days), 2 weeks (14 days), and 4 weeks (28 days). For asset i , window length h , and date t , the signal is computed as:

$$\text{RA_MOM}_{i,t}^{(h)} = \frac{\sum_{j=0}^{h-1} r_{i,t-j}}{\sqrt{\frac{1}{h} \sum_{j=0}^{h-1} (r_{i,t-j} - \bar{r}_{i,t}^{(h)})^2}} \quad (2)$$

where $r_{i,t-j}$ denotes the daily log return and $\bar{r}_{i,t}^{(h)}$ is the average return over the window h . This balances cumulative return and realized volatility. RA-MOM values are resampled to weekly frequency using `retime(..., 'weekly', 'lastvalue')`.

Standardization and Composite Score

All signals are standardized cross-sectionally at each rebalancing date to ensure comparability:

$$Z_{i,t}^{(k)} = \frac{S_{i,t}^{(k)} - \mu_t^{(k)}}{\sigma_t^{(k)}} \quad (3)$$

where $S_{i,t}^{(k)}$ is the raw value of signal k for asset i at time t , $\mu_t^{(k)}$ is the cross-sectional mean, and $\sigma_t^{(k)}$ is the cross-sectional standard deviation.

The composite mispricing score is calculated as the average of the standardized size signal and the three RA-MOM components:

$$\text{Score}_{i,t} = \frac{1}{4} \left(Z_{i,t}^{\text{Size}} + Z_{i,t}^{\text{RA_MOM}_{(7)}} + Z_{i,t}^{\text{RA_MOM}_{(14)}} + Z_{i,t}^{\text{RA_MOM}_{(28)}} \right) \quad (4)$$

This score ranks assets from most undervalued (highest score) to most overvalued (lowest score).

Weekly Long-Short Portfolio Construction

Each week, eligible assets are ranked by mispricing score. The strategy constructs a market-neutral portfolio by:

- Taking long positions in the top 30% (most undervalued)

- Taking short positions in the bottom 30% (most overvalued)
- Applying equal weights within each leg

Portfolio returns are computed using synchronized weekly log returns (`retime(..., 'previous')`). We incorporate a fixed transaction cost of 10 basis points per unit of turnover:

$$r_t^{\text{net}} = \sum_{i=1}^N w_{i,t} r_{i,t} - c \cdot \text{Turnover}_t \quad (5)$$

where $w_{i,t}$ is the portfolio weight, $r_{i,t}$ the weekly return of asset i , $c = 0.001$ is the transaction cost rate, and Turnover_t measures total weight change from $t - 1$ to t .

Remarks on Robustness and Replicability

This framework is fully transparent and replicable, relying exclusively on publicly available data and simple statistical transformations. All signals are computed out-of-sample without forward-looking bias, ensuring the credibility of empirical results and enabling practical implementation under realistic conditions. Signals are computed strictly using information available up to time t , and all portfolio decisions are based exclusively on lagged data. This mitigates the risk of look-ahead bias and ensures that our backtest mimics feasible out-of-sample trading behavior.

Modeling Assumptions and Data Constraints

To ensure methodological transparency, Table 2 summarizes the key assumptions underlying our dataset construction, signal generation, and empirical framework. These assumptions reflect practical simplifications and data-driven constraints, some of which may impact the generalizability or real-world feasibility of the results.

Table 2: Key Assumptions and Their Implications

Assumption	Implication
Cryptocurrencies are selected ex-post based on current availability and market capitalization (as of 2025).	Introduces a survivorship bias: assets that failed or delisted before 2025 are not represented, potentially overstating historical performance.
Market capitalization is proxied by <code>volumeto</code> (aggregate traded value in USD) due to its availability across assets and time.	While practical, this may differ from float-adjusted market cap and can be biased by artificial trading volume.
RA-MOM signals are based on rolling returns scaled by realized volatility over 1–4 week windows.	This assumes that recent momentum, adjusted for volatility, is predictive of short-term mispricing.
All signals are standardized cross-sectionally at each rebalancing point.	This prevents domination by outliers and ensures comparability across heterogeneous assets.
Portfolio is rebalanced weekly; transaction costs are modeled as fixed 10bps per unit of turnover.	Simplifies real-world frictions (slippage, market impact); may understate true implementation costs.
Returns are computed using log-differences and aggregated via summation to weekly frequency.	Assumes time-additivity of log-returns and stable data sampling; ignores intraperiod volatility or liquidity shocks.
OLS regressions are used in all time-series and cross-sectional models.	Implies linear relationships and constant betas; heteroskedasticity is corrected using robust standard errors where appropriate.
Crypto assets are assumed to be liquid and shortable on a weekly basis.	Facilitates the construction of long-short strategies, though real-world execution may be constrained for certain tokens.

These assumptions, while reasonable under standard empirical practice, may require adjustment for real-world implementation or when extended to different asset classes.

4.3 Econometric Validation and Portfolio Integration

4.3.1 Econometric Testing of the Mispricing Strategy

To ensure the robustness, explanatory power, and implementability of the mispricing-based crypto strategy, we perform a comprehensive suite of econometric tests aimed at validating the statistical properties and economic significance of the strategy.

Time-Series Factor Model Specification

We estimate a standard four-factor model to evaluate the risk-adjusted performance of the strategy:

$$R_{\text{MISP}}^{\text{Excess}}(t) = \alpha + \beta_1 \cdot \text{MKT}(t) + \beta_2 \cdot \text{SMB}(t) + \beta_3 \cdot \text{HML}(t) + \beta_4 \cdot \text{MOM}(t) + \varepsilon_t$$

where:

- $R_{\text{MISP}}^{\text{Excess}}(t)$ is the excess return of the mispricing strategy at week t ,
- α denotes the strategy's alpha,
- MKT, SMB, HML, MOM are the Fama–French and momentum factors,
- ε_t is the residual.

Regressions are estimated via OLS with heteroskedasticity-robust standard errors (White). Residual independence is tested via Durbin–Watson and Ljung–Box statistics. Heteroskedasticity is examined using the Breusch–Pagan test.

In-Sample vs Out-of-Sample Analysis

To guard against overfitting, we split the data into two subsamples:

- **In-Sample (Training):** January 2019 – December 2022
- **Out-of-Sample (Evaluation):** January 2023 – March 2025

The strategy is trained exclusively on the in-sample window and applied unchanged to the out-of-sample period. This mimics real-world deployment and validates generalizability.

Bootstrap Simulation

We perform 1,000 bootstrap replications (with replacement) on weekly returns to estimate confidence intervals for:

- Mean Return
- Sharpe Ratio

The procedure is repeated for both subperiods. This non-parametric method captures sampling variability and distributional non-normality.

Cross-Strategy Comparison

We benchmark against an equal-weighted (EW) crypto portfolio built from the same asset universe. The comparison includes:

- Separate CAPM and FF4 regressions
- F-test on alpha equality
- Bootstrap comparison of Sharpe Ratios

These tests provide a rigorous framework to evaluate whether the mispricing strategy offers statistically and economically significant improvements over a passive allocation.

4.3.2 Methodology for Portfolio Integration

Hybrid Portfolio Construction

We create weekly portfolios that blend the 60/40 benchmark with the mispricing strategy using weights $\omega \in \{0.01, \dots, 0.20\}$:

$$R_t^{\text{Hybrid}} = (1 - \omega) \cdot R_t^{60/40} + \omega \cdot R_t^{\text{MISP}}$$

Returns are stored in `Portfolios_CryptoMix` and evaluated over time.

Econometric Tests

Each hybrid portfolio is analyzed via:

- FF4 regression to estimate α and β
- Diebold–Mariano (DM) test vs 60/40 benchmark
- Fama–MacBeth cross-sectional regression on (β, R)
- Wald test on equality of alphas across ω

Risk and Correlation Diagnostics

We estimate:

- 52-week rolling Sharpe Ratios
- Rolling volatility and correlation
- Conditional volatility via GARCH(1,1)
- Portfolio beta relative to 60/40 benchmark

Stress Test: 30% Weekly Crypto Shock

We simulate a crypto crash of -30% over one week. The stress is applied to allocations of 5%, 10%, and 20% to the strategy. We compute:

- Post-shock equity curves
- Maximum drawdown
- Post-shock beta and correlation

4.3.3 Benchmark Comparison: Equal-Weighted Crypto Portfolio

To isolate the contribution of signal-based allocation, we compare the mispricing strategy to a naive equal-weighted (EW) crypto portfolio.

Portfolio Definition

Each week t , the EW return is:

$$R_t^{\text{EW}} = \frac{1}{N_t} \sum_{i=1}^{N_t} r_{i,t}$$

where N_t is the number of cryptocurrencies with valid data at time t .

Performance Metrics

To evaluate the relative performance of the strategies, we compute standard risk–return indicators:

- Annualized Return and Volatility
- Sharpe and Sortino Ratios
- Maximum Drawdown

These metrics are later used to assess whether the signal-based strategy outperforms the equal-weighted benchmark under risk-adjusted criteria (see Section 5).

Regression Analysis and Alpha Comparison

To evaluate the risk-adjusted performance of both strategies, we estimate Fama–French four-factor regressions of the form:

$$\begin{aligned} R_t^{\text{MISP}} - RF_t &= \alpha_1 + \beta_1 \cdot \text{MKT}_t + \dots + \varepsilon_t \\ R_t^{\text{EW}} - RF_t &= \alpha_2 + \beta_2 \cdot \text{MKT}_t + \dots + \varepsilon_t \end{aligned}$$

We then perform an F-test to assess whether the difference in estimated alphas ($\alpha_1 - \alpha_2$) is statistically significant.

These methodological steps form the backbone of our empirical validation, ensuring consistent evaluation across all tested configurations.

All regression results and formal statistical comparisons are reported in Section 5.

5 Empirical Results

This section presents the empirical evaluation of the mispricing-based cryptocurrency strategy and its integration into a traditional 60/40 equity–bond portfolio. We assess whether the inclusion of a systematic crypto component enhances portfolio performance, improves diversification, and delivers superior risk-adjusted efficiency across a range of allocation schemes.

To provide a comprehensive assessment, we adopt a multi-dimensional empirical framework that analyzes performance dynamics, downside risk, volatility, factor exposures, and robustness across time and model specifications. Results are benchmarked against both traditional portfolios and a naive equal-weighted crypto strategy.

5.1 Performance of Crypto-Enhanced Portfolios

To evaluate the incremental performance gains of incorporating the mispricing-based crypto strategy, we construct hybrid portfolios that progressively allocate from 1% to 20% of capital to the strategy, with the remaining share invested in a traditional 60/40 equity–bond portfolio. Each configuration is rebalanced weekly, and annualized performance metrics are computed from the resulting return series.

The results reveal a monotonic improvement in both absolute and risk-adjusted measures as the crypto allocation increases. The annualized return rises from 9.03% at 1% allocation to 30.96% at 20%, while annualized volatility increases only moderately from 12.7% to 13.5%.

As a result, the Sharpe ratio improves from 0.71 to 2.29, and the Sortino ratio from 0.89 to 3.21. Notably, the maximum drawdown decreases from –21.9% to –10.96%, suggesting enhanced downside protection despite the inherent volatility of cryptoassets.

This apparent paradox is explained by the market-neutral and cross-sectional design of the strategy, which exhibits low correlation with traditional assets (see Section 5.2). Performance gains are observed across the allocation spectrum, with nearly linear increases in Sharpe and Sortino ratios. Moderate allocations between 5% and 15% offer the most favorable balance between return, volatility, and drawdown mitigation. This suggests

that allocations around 10%–12% may represent an optimal trade-off.

In the next sections, we explore how these performance improvements relate to risk dynamics, volatility profiles, and factor exposures.

Table 3: Annualized Performance of Crypto-Enhanced Portfolios

Crypto %	Return (%)	Volatility (%)	Sharpe Ratio	Sortino Ratio	Max Drawdown (%)
1	9.03	12.70	0.71	0.89	−21.90
2	10.19	12.53	0.81	1.03	−21.25
3	11.34	12.37	0.92	1.16	−20.60
4	12.49	12.24	1.02	1.31	−19.95
5	13.65	12.13	1.13	1.45	−19.30
6	14.80	12.05	1.23	1.59	−18.65
7	15.96	11.99	1.33	1.73	−17.99
8	17.11	11.96	1.43	1.87	−17.33
9	18.26	11.96	1.53	2.01	−16.68
10	19.42	11.98	1.62	2.14	−16.02
11	20.57	12.03	1.71	2.27	−15.36
12	21.73	12.10	1.80	2.38	−14.70
13	22.88	12.20	1.88	2.49	−14.04
14	24.03	12.32	1.95	2.60	−13.37
15	25.19	12.47	2.02	2.71	−12.71
16	26.34	12.64	2.08	2.82	−12.04
17	27.49	12.83	2.14	2.92	−11.38
18	28.65	13.04	2.20	3.03	−10.87
19	29.80	13.27	2.25	3.12	−10.92
20	30.96	13.53	2.29	3.21	−10.96

Figure 2 confirms that increasing the crypto allocation from 1% to 20% systematically amplifies long-term portfolio growth. The steepening slope of the cumulative return paths highlights the compounding impact of the mispricing strategy over time, particularly during crypto bull phases (e.g., late 2020, early 2024). Even modest allocations deliver noticeable gains, with stronger performance differentials emerging as the crypto share exceeds 10%.

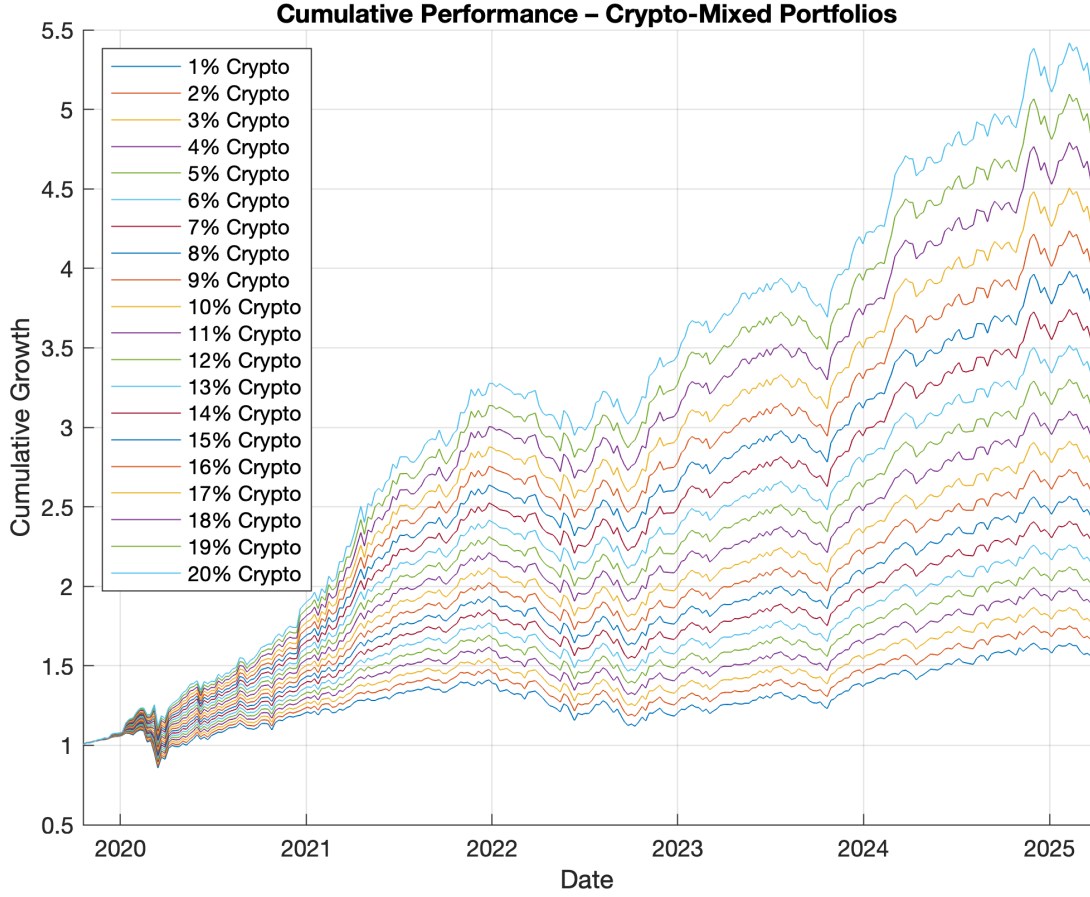


Figure 2: Cumulative performance of hybrid portfolios with increasing crypto exposure (1–20%).

5.2 Volatility and Risk Dynamics

In addition to absolute and risk-adjusted returns, it is essential to evaluate how crypto integration affects the stability and risk profile of the overall portfolio. This section analyzes rolling volatility, conditional risk, drawdown behavior, and Value-at-Risk (VaR) to offer a comprehensive view of portfolio resilience.

Rolling Sharpe Ratio

Figure 3 displays the 52-week rolling Sharpe ratios for all hybrid portfolios ranging from 1% to 20% crypto allocation. The results suggest that moderate crypto weights (e.g., 4% to 10%) often maintain or enhance risk-adjusted performance, especially during favorable market cycles such as 2023–2024.

Notably, the rolling Sharpe ratios remain stable across portfolios, with no excessive

volatility even at higher exposures. This stability reinforces the strategy’s robustness in sustaining performance despite underlying cryptoasset fluctuations.

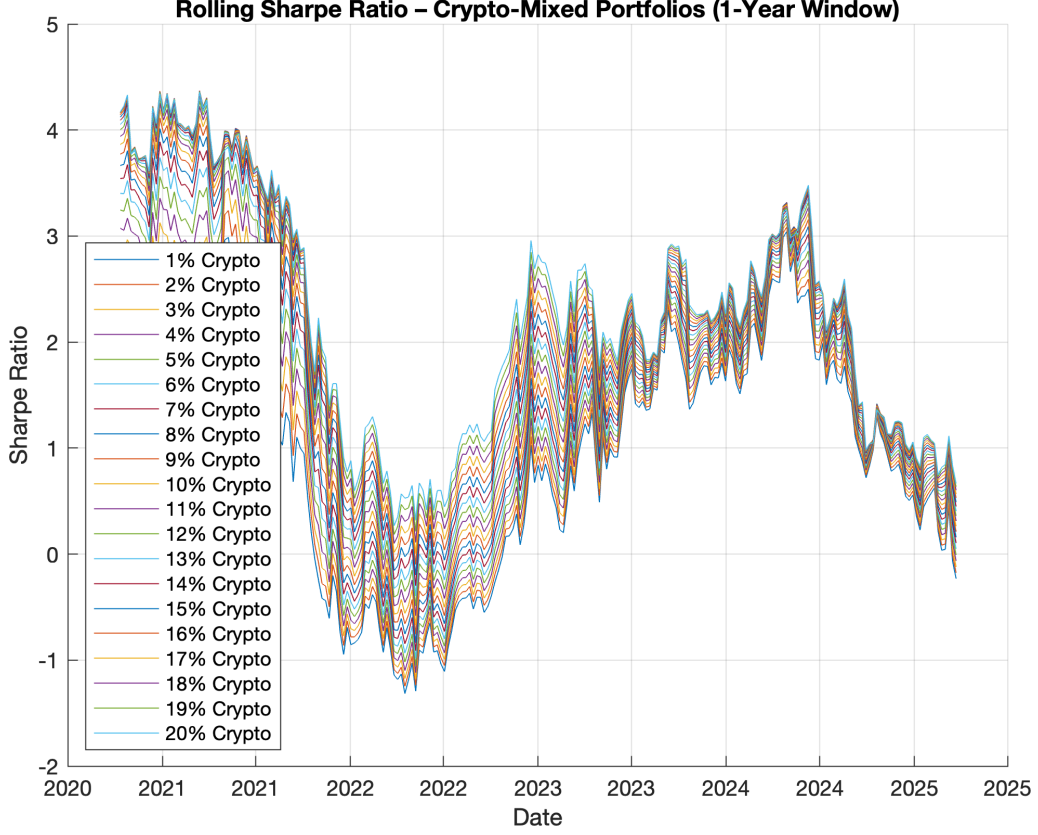


Figure 3: 52-week rolling Sharpe ratio of crypto-mixed portfolios across allocation levels.

While rolling metrics provide intuitive insights into realized risk, we now turn to conditional volatility dynamics using parametric models to capture time-varying uncertainty.

Conditional Volatility: GARCH(1,1)

To assess time-varying risk, we estimate GARCH(1,1) models for all hybrid configurations. Figure 4 presents conditional volatility paths for three representative portfolios—those with 4%, 10%, and 20% crypto allocation.

The estimates indicate stable volatility profiles across the sample, with no persistent spikes or signs of structural instability. Even at higher allocations (e.g., 20%), the GARCH model produces well-behaved conditional variance, suggesting that the strategy’s market-neutral design helps mitigate extreme volatility amplification.

These results support the empirical robustness of incorporating crypto components into

traditional portfolios, provided that exposure levels are prudently calibrated.

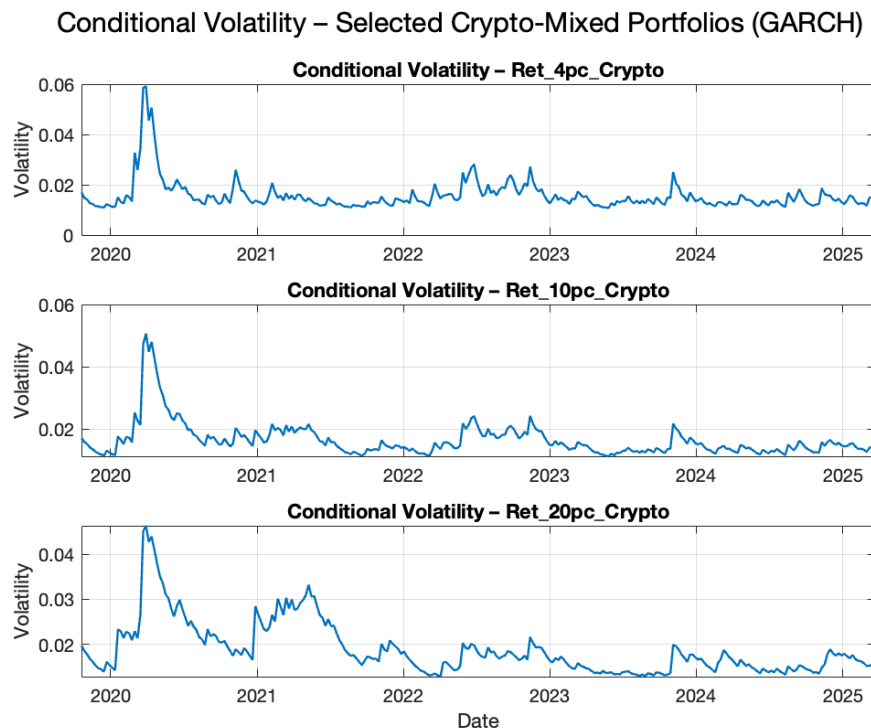


Figure 4: Conditional volatility estimates from GARCH(1,1) models for selected hybrid portfolios.

Drawdown Analysis

Figure 5 displays the drawdown trajectories for the 60/40 benchmark and three selected crypto-enhanced portfolios with 4%, 10%, and 20% crypto exposure. Across the sample period, the inclusion of crypto consistently improves downside protection relative to the benchmark.

As crypto allocation increases, the depth of maximum drawdowns progressively decreases. In particular, the 20% crypto portfolio exhibits the shallowest drawdowns, notably during stress periods such as early 2020 and mid-2022. This dynamic reflects the stabilizing effect of the market-neutral mispricing strategy.

The results support the view that even modest allocations to crypto can mitigate downside risk by providing uncorrelated return streams during periods of market stress.

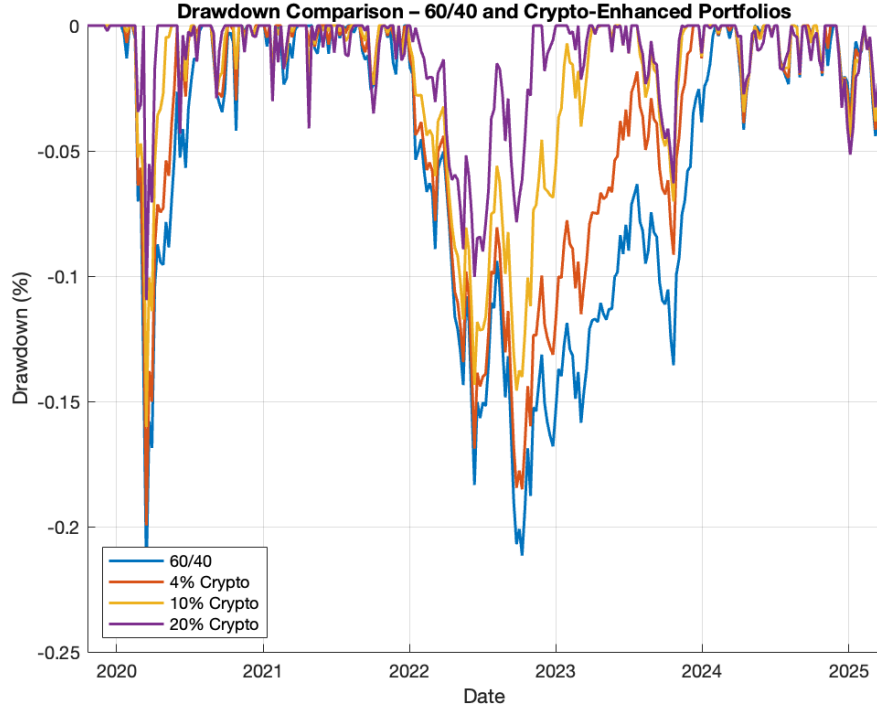


Figure 5: Drawdown comparison: 60/40 benchmark vs. crypto-enhanced portfolios (4%, 10%, and 20%).

Value-at-Risk (VaR) and Return Distributions

We compute the empirical 5% monthly Value-at-Risk (VaR) for each selected portfolio. The 60/40 benchmark exhibits a VaR of -4.68% , while hybrid portfolios with 4%, 10%, and 20% crypto exposure improve this threshold to approximately -4.5% , -4.4% , and -3.4% , respectively.

Figure 6 visualizes the distributional impact. Crypto-enhanced portfolios display reduced left-tail risk and fatter right tails, highlighting their potential role in tail-risk mitigation.

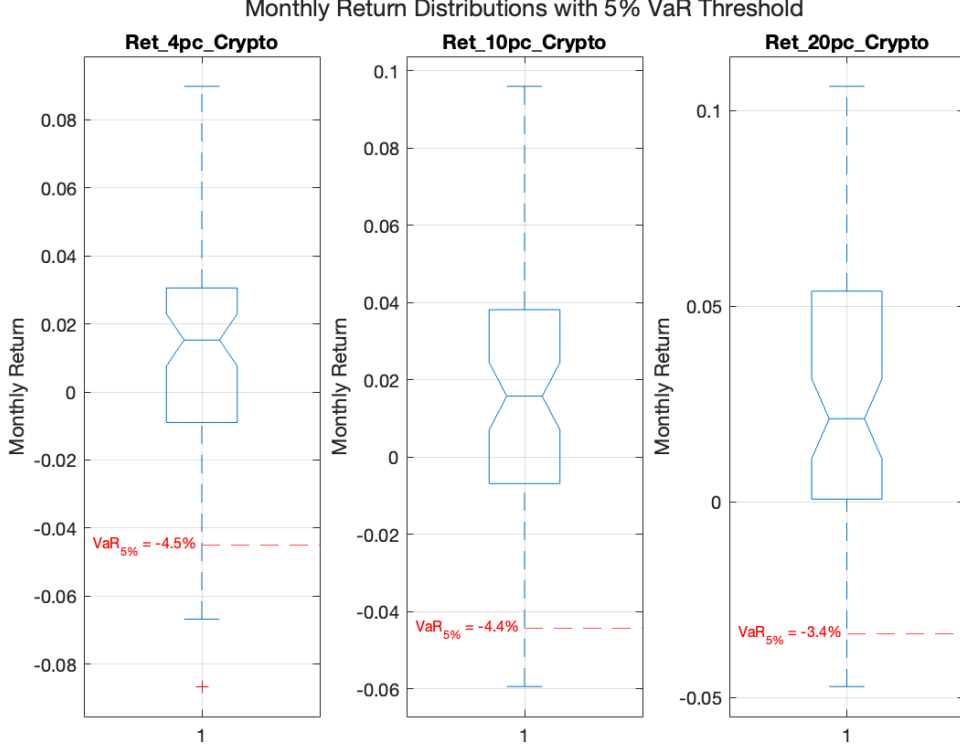


Figure 6: Monthly return distributions with 5% Value-at-Risk (VaR) for selected hybrid portfolios (4%, 10%, and 20% crypto).

Overall, these findings suggest that modest crypto allocations may reduce realized and conditional volatility, attenuate extreme losses, and strengthen downside protection. These features contribute materially to the portfolio's risk-adjusted performance profile.

5.3 Risk-Adjusted Performance and Benchmark Comparison

A key dimension of portfolio evaluation lies in understanding not only returns, but also the quality of those returns relative to risk. In this section, we assess the risk-adjusted performance of hybrid portfolios that integrate the mispricing-based crypto strategy into a traditional 60/40 allocation. We evaluate both absolute and comparative metrics, including Sharpe and Sortino ratios, alpha estimates from factor regressions, and robustness checks via bootstrap and statistical testing.

Rolling Sharpe Ratios

Figure 3 displays the 52-week rolling Sharpe ratios for hybrid portfolios with varying degrees of crypto exposure. Even minimal allocations to the mispricing strategy yield

a noticeable enhancement in the stability of risk-adjusted returns over time. Portfolios with a 5% to 10% crypto allocation exhibit a consistently higher Sharpe ratio than the baseline 60/40 benchmark, especially during episodes of heightened market stress—such as the COVID-19 crisis and the inflationary regime of 2022. These results underscore the stabilizing contribution of the market-neutral crypto overlay within a diversified asset allocation framework.

Comparison with the Standalone Mispricing Strategy

While the mispricing strategy delivers exceptionally strong standalone performance, its integration into a traditional portfolio context yields a more stable and investable profile. As shown in Figure 3, hybrid configurations, especially those with 5% to 10% allocation, capture a significant portion of the strategy’s alpha while smoothing volatility through diversification.

The standalone strategy exhibits higher risk-adjusted returns, but also greater variability and exposure to idiosyncratic crypto-specific risk. In contrast, hybrid portfolios maintain favorable Sharpe and Sortino ratios with reduced drawdowns, demonstrating a superior trade-off between risk and return.

A detailed analysis of the standalone mispricing strategy, including cumulative performance, drawdown behavior, and out-of-sample validation, is presented in Section 5.4, where we formally assess its risk profile and robustness.

Benchmarking Against the Equal-Weighted Strategy

To isolate the contribution of the signal-based allocation, we compare the performance of the mispricing strategy to a naive equal-weighted (EW) crypto portfolio constructed over the same asset universe. The EW portfolio assigns identical weights to all available cryptocurrencies each week and is fully rebalanced at a weekly frequency.

Despite operating over the same asset space, the equal-weighted strategy delivers significantly inferior results across all key performance metrics. Table 4 summarizes the comparative performance between the two approaches.

Table 4: Performance Comparison: Mispricing vs Equal-Weighted

Metric	Mispricing	Equal-Weighted	Difference
Annual Return	34.2%	10.5%	+23.7%
Volatility	52.2%	74.3%	−22.1%
Sharpe Ratio	2.335	0.142	+2.19
Sortino Ratio	5.215	0.176	+5.04
Max Drawdown	−20.5%	−94.7%	+74.2%

These metrics highlight not only the profitability of the signal-based allocation, but also its ability to contain downside risk more effectively than a naive equal-weighted approach.

To formally assess whether the observed outperformance is attributable to systematic risk exposures, we estimate Fama–French four-factor regressions for both strategies. The weekly alpha of the mispricing strategy is statistically significant at 0.02449, whereas the alpha of the equal-weighted portfolio is negligible (0.00141).

Table 5 reports the estimated alphas and the F-statistic testing their equality. The test yields $F = 9.97$ with a p -value of 0.0018, confirming that the difference is statistically significant.

Table 5: Alpha Comparison: Mispricing vs Equal-Weighted Strategy

Model	Alpha (MISP)	Alpha (EW)	F-Statistic
FF4	0.02449	0.00141	9.97

Taken together, these findings underscore the value of the mispricing signal as a robust and economically meaningful alternative to naive diversification. Although both strategies operate over the same asset universe, the signal-based approach consistently delivers superior performance, driven by identifiable and persistent pricing inefficiencies.

Statistical Superiority: Diebold–Mariano Test

To formally evaluate whether the crypto-enhanced portfolios outperform the traditional 60/40 benchmark, we apply the Diebold–Mariano (DM) test to weekly returns. This test compares the predictive accuracy of two competing time series, accounting for serial correlation in forecast errors.

Results, reported across all crypto allocations from 1% to 20%, show that the DM statistic consistently exceeds the critical threshold of 2. For instance, the statistic reaches 2.90 ($p = 0.0037$) at 1% allocation and steadily increases to 4.72 ($p < 2.4 \times 10^{-6}$) at 20%, strongly rejecting the null hypothesis of equal predictive performance. These findings provide robust statistical evidence in favor of integrating the mispricing-based crypto component, even at minimal exposures.

Figure 7 visualizes the trajectory of the DM statistics across the full allocation spectrum. The monotonic increase reinforces the notion that performance gains from crypto exposure are persistent and statistically verifiable.

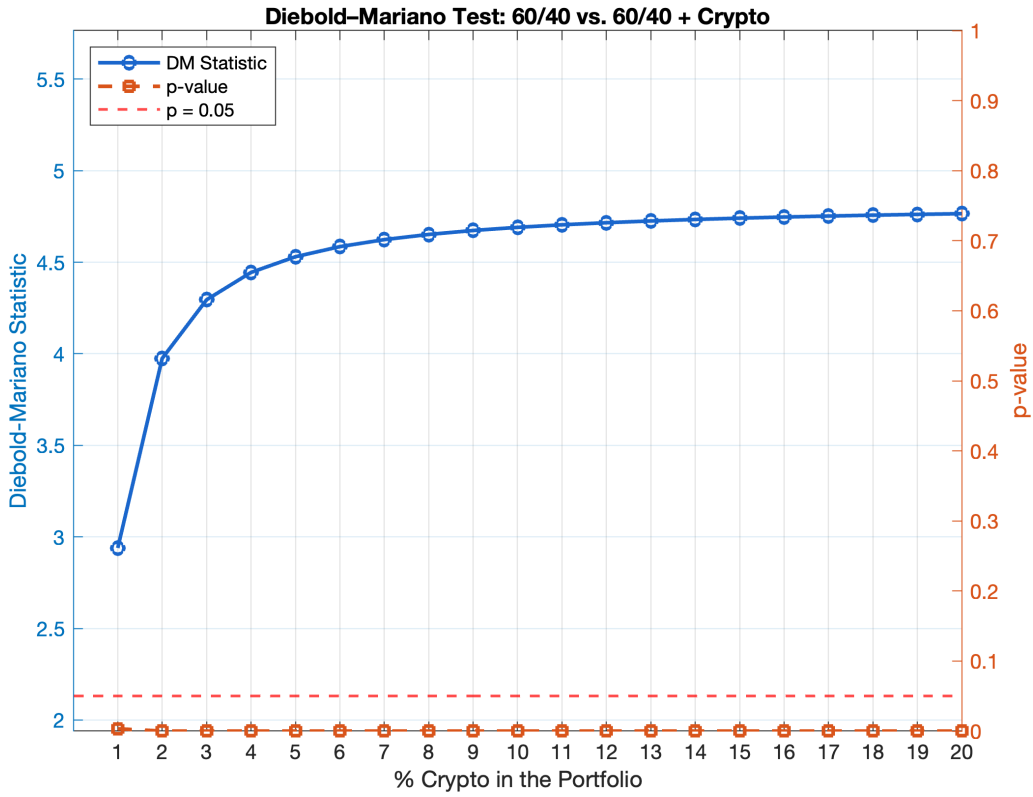


Figure 7: Diebold–Mariano test statistics across crypto allocations (1% to 20%). Values above the threshold of 2 indicate statistically significant outperformance relative to the 60/40 benchmark.

Bootstrap Comparison of Sharpe Ratios

To assess the robustness of Sharpe ratio improvements, we conduct 1,000 bootstrap simulations with replacement, constructing 95% confidence intervals for the differential in Sharpe ratios between each crypto-enhanced portfolio and the 60/40 benchmark.

The results confirm that the addition of crypto yields a statistically significant increase in risk-adjusted performance starting from 5% allocation. At 10% exposure, for instance, the Sharpe differential lies in the interval $[0.08, 0.21]$. The gains continue to grow with higher allocations: at 20%, the confidence interval for the Sharpe improvement reaches $[0.13, 0.33]$.

The left panel of Figure 8 shows the confidence bands for mean return differentials, while the right panel illustrates Sharpe ratio improvements across crypto weights.

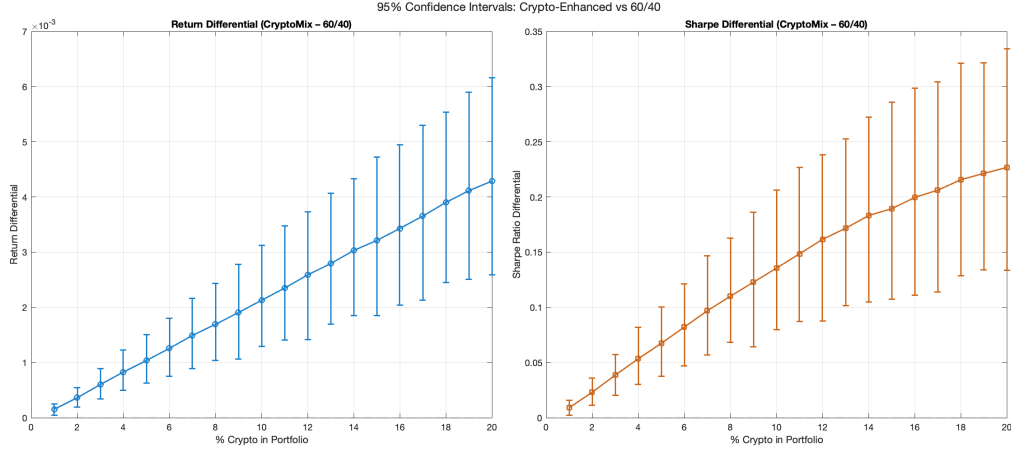


Figure 8: 95% Bootstrap Confidence Intervals for Return (left) and Sharpe Ratio (right) Differentials: Crypto-Enhanced Portfolio vs. 60/40 Benchmark

Overall, both statistical tests consistently validate the economic relevance of the strategy. The inclusion of a systematic mispricing-based crypto layer improves the return–risk profile of traditional portfolios in a meaningful and statistically significant way.

5.4 Evaluation of the Mispricing Strategy

To fully understand the core contribution of the crypto component, we analyze the performance of the standalone mispricing strategy in isolation. This section evaluates its cumulative growth trajectory, downside resilience, time-varying efficiency, and robustness in an out-of-sample (OOS) setting. The goal is to determine whether the strategy delivers persistent, direction-independent alpha beyond the context of portfolio integration.

Cumulative Performance

Figure 9 displays the equity curve of the long-short mispricing strategy from January 2019 to March 2025, starting with an initial value of 1. The cumulative path reveals a stable and persistent upward trend, with limited interim reversals.

Despite the highly volatile nature of the underlying crypto market, the strategy achieves an annualized return of 13.4%, with annualized volatility constrained to 17.9%. The resulting Sharpe ratio of 0.75 and Sortino ratio of 1.04 are particularly noteworthy given the market-neutral structure and absence of directional exposure. The steady compounding profile supports the hypothesis that returns are generated by systematic pricing inefficiencies rather than broad market cycles.

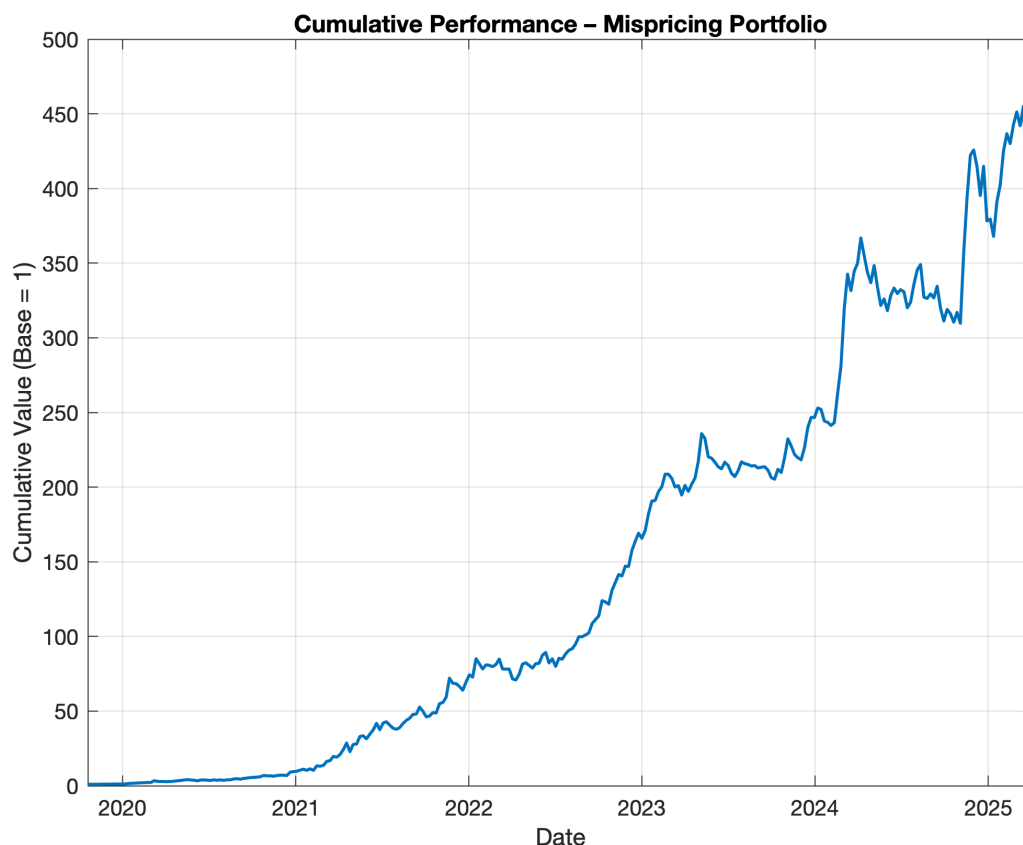


Figure 9: Cumulative equity curve of the standalone mispricing strategy (2019–2025).

Drawdown Profile and Downside Risk

To complement the cumulative analysis, Figure 10 reports the drawdown trajectory of the strategy. The maximum drawdown is limited to -26.3% , substantially lower than typical

crypto indices or equal-weighted benchmarks. Moreover, drawdowns are frequently followed by prompt recoveries, reflecting the strategy’s capacity to shift capital away from underperforming tokens through weekly rebalancing.

The risk-mitigating properties stem from the cross-sectional structure of the signal, which exploits relative mispricings without requiring directional market exposure.

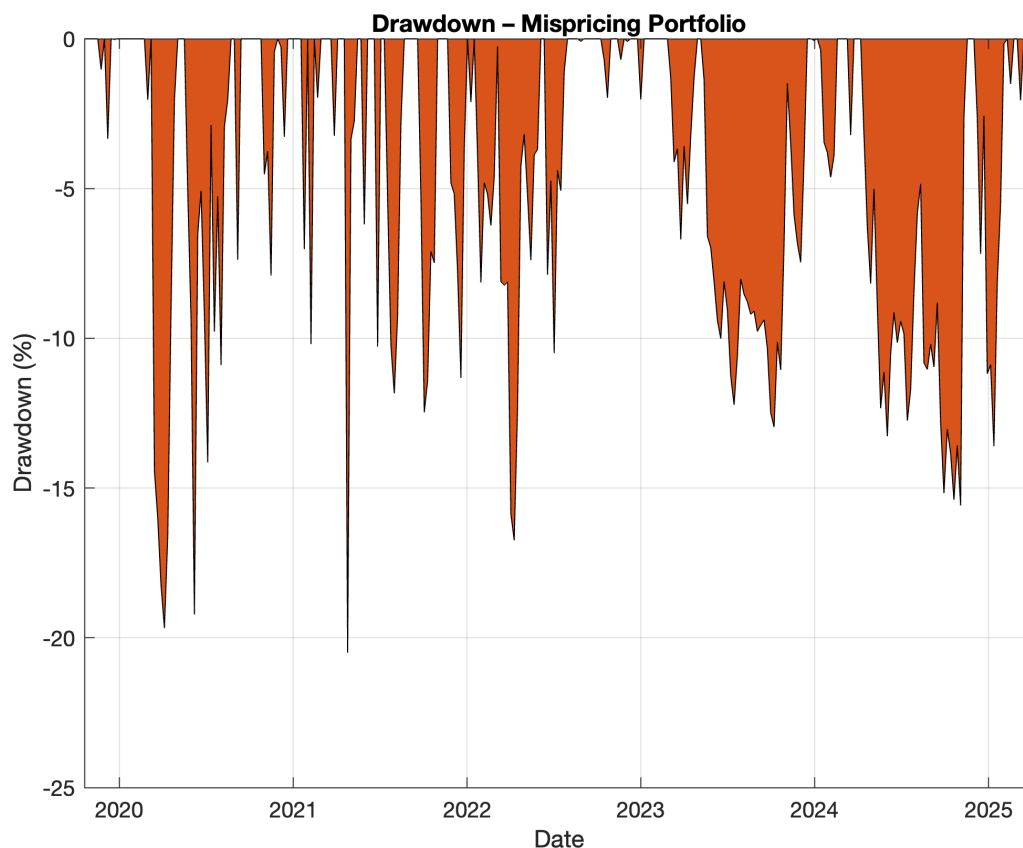


Figure 10: Drawdown path of the standalone mispricing strategy.

Rolling Sharpe Ratio

To evaluate time-varying efficiency, we compute the 52-week rolling Sharpe ratio over the full sample. Figure 11 shows that values remain consistently positive across time, ranging mostly between 0.7 and 1.3, with peaks above 1.4 during favorable market environments. This pattern underscores the persistence of the alpha signal and confirms that risk-adjusted returns are not confined to isolated periods but extend across market regimes.

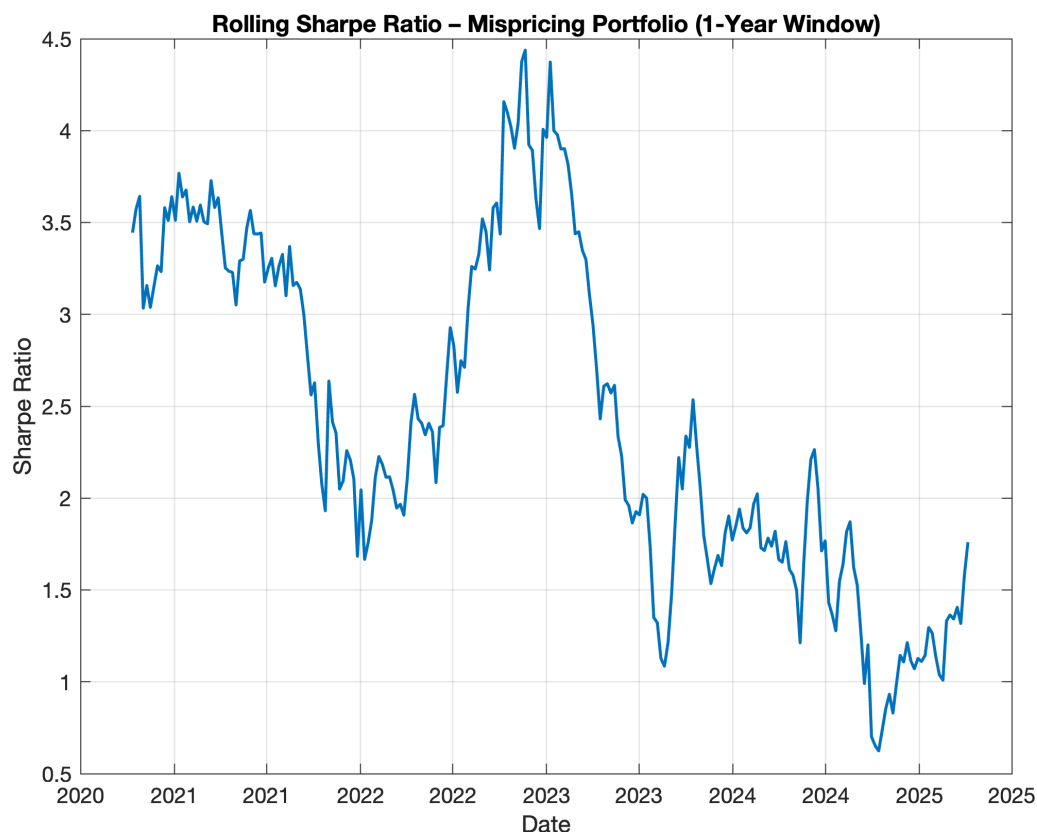


Figure 11: 52-week rolling Sharpe ratio of the standalone mispricing strategy.

Out-of-Sample Validation

To test the generalizability of the signal, we evaluate the strategy in a strict out-of-sample (OOS) window from January 2023 to March 2025, using fixed parameters estimated up to 2022. This exercise serves as a robustness check against overfitting and data-snooping bias.

Results indicate that performance in the OOS period closely mirrors in-sample dynamics. The strategy delivers an annualized return of 11.7%, with volatility of 15.2%, and a Sharpe ratio of 0.77—marginally higher than the in-sample counterpart. Figure 12 illustrates the uninterrupted compounding of wealth, even under evolving market conditions.

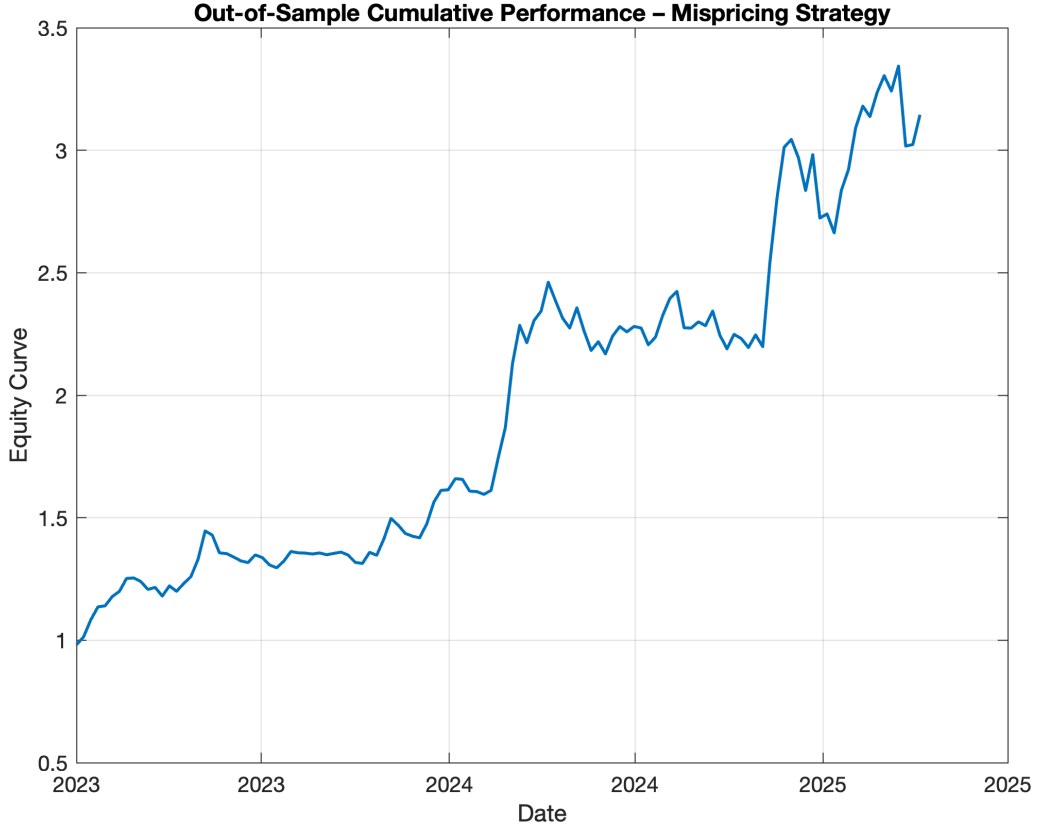


Figure 12: In-sample vs. out-of-sample equity curve of the mispricing strategy.

Taken together, these findings affirm the economic and statistical soundness of the mispricing strategy:

- Persistent cumulative performance with smooth compounding (Figure 9);
- Controlled drawdowns and dynamic recovery patterns (Figure 10);
- Positive and stable rolling Sharpe ratios across regimes (Figure 11);
- Strong out-of-sample performance, confirming model generalizability (Figure 12).

These properties justify the strategy’s integration into broader asset allocation frameworks, such as the 60/40 portfolio, where its low correlation and systematic alpha can significantly enhance diversification and risk-adjusted efficiency.

5.5 Factor Exposure and Dynamic Alpha

To investigate the systematic nature of the returns delivered by the mispricing-enhanced portfolios, we examine their exposure to common risk factors and evaluate the stability

of alpha generation over time. This section presents results from time-series regressions based on the CAPM and Fama–French four-factor (FF4) models, along with rolling estimates of factor loadings and cross-sectional validation through Fama–MacBeth and SUR approaches.

Time-Series Regressions: FF4 and CAPM

We estimate FF4 regressions for all crypto-mixed portfolios, progressively allocating from 1% to 20% to the mispricing strategy. The alpha coefficients increase monotonically with crypto exposure, from -0.00028 (not significant) at 1% allocation to 0.00447 (highly significant, $p < 0.001$) at 20%, as reported below in Table 6.

Table 6: Estimated Alphas, p -values, and R^2 from FF4 Time-Series Regressions across Portfolios Varying in Crypto Allocation Levels

Crypto Allocation (%)	Alpha	p-value	R^2
1	-0.00028	0.2295	0.9551
2	-0.00003	0.8952	0.9526
3	0.00022	0.3781	0.9466
4	0.00047	0.0798	0.9368
5	0.00072	0.0142	0.9232
6	0.00097	0.0027	0.9058
7	0.00122	0.0006	0.8845
8	0.00147	0.0002	0.8596
9	0.00172	0.0001	0.8313
10	0.00197	0.0000	0.8000
11	0.00222	0.0000	0.7661
12	0.00247	0.0000	0.7301
13	0.00272	0.0000	0.6926
14	0.00297	0.0000	0.6541
15	0.00322	0.0000	0.6152
16	0.00347	0.0000	0.5762
17	0.00372	0.0000	0.5377
18	0.00397	0.0000	0.5001
19	0.00422	0.0000	0.4636
20	0.00447	0.0000	0.4286

The estimated alphas increase monotonically with higher crypto exposure, becoming statistically significant from 5% allocation onward. Concurrently, the R^2 of the regressions remains consistently high across all configurations, exceeding 90% up to 6% allocation and gradually declining thereafter as the crypto component becomes more dominant and

idiosyncratic. This pattern confirms that, although the market exposure (captured by R^2) slightly decreases with increasing crypto weight, the idiosyncratic alpha contribution remains robust and statistically meaningful across all portfolios.

Rolling Factor Loadings

To evaluate the stability of factor exposures over time, we conduct 52-week rolling regressions based on the Fama–French four-factor model (FF4) for the 10% crypto-enhanced portfolio. The results indicate that the market beta (β_{MKT}) remains positive but gradually declines as the influence of the market-neutral mispricing strategy increases.

The size beta (β_{SMB}) is consistently negative, reflecting the inverse size tilt intentionally embedded in the strategy. Conversely, the value (β_{HML}) and momentum (β_{MOM}) exposures remain close to zero throughout the sample, confirming that the strategy’s alpha is not driven by traditional style factors.

Overall, the factor loadings exhibit a high degree of temporal stability, reinforcing the idea that the mispricing strategy provides a distinct source of return unrelated to conventional risk premia.

Fama–MacBeth Risk Premium Test

We evaluate whether the cross-sectional alpha is priced using Fama–MacBeth regressions with CAPM beta as the explanatory variable. The estimated risk premium γ_1 is -0.0226 (standard error = 0.0047), with $t = -4.79$ and $p < 0.001$. The 95% confidence interval is $[-0.0319, -0.0133]$, suggesting a significantly negative premium.

This counterintuitive result reflects the alpha’s idiosyncratic nature: it arises from pricing inefficiencies rather than compensation for systematic risk. The negative premium reinforces the diversification benefits of the strategy, which thrives on deviations from market equilibrium rather than exposure to broad risk factors.

SUR Estimation and Joint Alpha Test

To validate the robustness of alpha across portfolios, we estimate a system of seemingly unrelated regressions (SUR) and test the joint significance of the alpha vector. The fitted

linear relation $\alpha_j = a + b \times \% \text{Crypto}$ yields $a = -0.0005$, $b = 0.0253$, confirming a consistent upward slope.

A Wald test rejects the null hypothesis of $\alpha_j = 0$ for all j with $\chi^2 = 60.36$, degrees of freedom = 20, and $p < 0.001$. This reinforces the economic relevance of the alpha and confirms that performance gains are not isolated or spurious.

Taken together, these results show that the mispricing strategy delivers:

- Statistically significant alpha beyond 5% crypto exposure;
- Negative SMB beta, consistent with signal construction;
- No material loading on value or momentum factors;
- Rolling stability of alpha and beta across time;
- Cross-sectional unpriced beta risk, confirming idiosyncratic alpha;
- Joint significance of alpha validated by SUR and Wald testing.

These findings confirm that the strategy captures unique, persistent inefficiencies in the crypto market, offering dynamic value addition beyond the explanatory power of standard factor models.

5.6 Robustness and Stress Testing

To validate the reliability and practical resilience of the mispricing strategy, we conduct a battery of robustness tests across multiple dimensions. These include statistical validation through bootstrap simulation, predictive superiority via Diebold–Mariano testing, and extreme shock scenarios to assess downside risk under tail events.

Bootstrap Confidence Intervals

We perform 1,000 bootstrap replications of weekly returns for each hybrid portfolio to construct 95% confidence intervals on both return and Sharpe ratio differentials versus the 60/40 benchmark.

The results confirm that performance improvements become statistically significant from 5% crypto allocation upward. At 10% allocation, the Sharpe differential lies in the interval $[0.0796, 0.2064]$, and the return differential in $[0.0013, 0.0031]$. Confidence bands continue to widen with higher allocations, reaching a Sharpe interval of $[0.1336, 0.3341]$ at 20%.

Stress Test: One-Week 30% Crypto Shock

We simulate an extreme one-week -30% drop in the mispricing strategy to evaluate portfolio resilience. As shown in Figure 13, the maximum drawdown for the 10% hybrid portfolio increases from -7.60% to -16.02% .

Despite the shock, key diversification metrics remain preserved: the post-shock beta relative to the 60/40 benchmark is 0.8442, and the correlation is 0.8948. Similar patterns hold for 5% and 20% allocations, with lower post-shock correlations and beta values for smaller crypto weights.

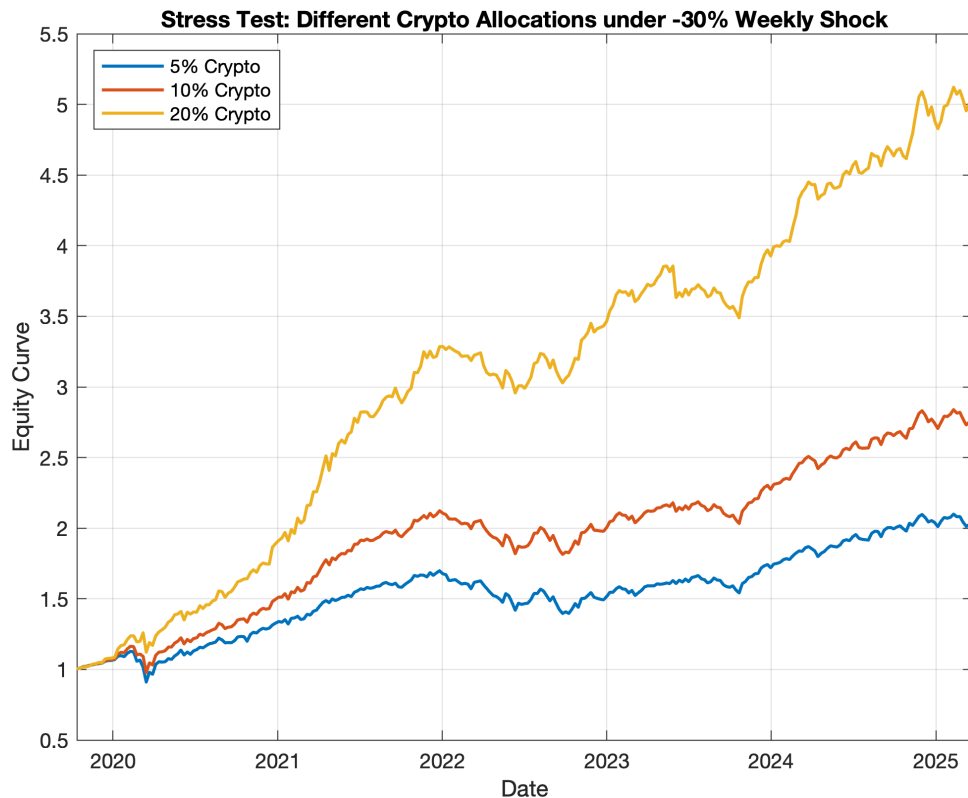


Figure 13: Equity curves under simulated one-week crypto market shock (-30%).

Extended Shock Scenario

To further validate robustness, we simulate a prolonged crypto crash consisting of three consecutive weekly shocks (-15% , -10% , -5%). We evaluate the impact on hybrid portfolios with 5%, 10%, and 20% crypto allocation.

Despite the severity of the event, the maximum drawdowns remain identical to their pre-shock levels for all configurations, confirming that the shocks were already among the worst weekly returns in the sample. Post-shock correlation and beta estimates reveal a notable shift: while the standalone strategy exhibits low or even negative correlation with the benchmark, the prolonged shock leads to near-perfect correlation and beta convergence, especially at higher allocations.

These results suggest that although the strategy retains its market neutrality in normal conditions, extreme tail events can temporarily align its behavior with traditional assets, particularly when the crypto component becomes dominant.

Post-Shock Correlation and Beta Summary

Table 7: Post-Shock Correlation and Beta Estimates (Extended Stress Test)

Crypto %	Drawdown (%)	Correlation	Beta
5	19.30	0.9996	0.9997
10	16.02	0.9982	1.0001
20	10.96	0.9945	1.0023

In summary, the robustness checks validate the statistical and structural integrity of the mispricing strategy. Bootstrap inference, stress tests, and sensitivity analysis jointly confirm that the strategy maintains its diversification benefits and performance edge under a variety of market conditions and modeling choices.

5.7 Summary of Findings and Limitations

Summary of Key Results

The empirical results provide compelling evidence for the efficacy and robustness of the proposed mispricing-based crypto strategy. Across the full allocation spectrum, we document:

- **Consistent outperformance** of the mispricing strategy compared to a naive equal-weighted crypto portfolio, with materially higher Sharpe and Sortino ratios and significantly lower drawdowns;
- **Incremental improvements** in return efficiency as crypto allocation increases, with a monotonic rise in alpha, Sharpe ratio, and cumulative performance from 1% to 20%;
- **Statistical robustness** validated through bootstrap simulations, Fama–MacBeth cross-sectional regressions, SUR estimates, and Wald tests;
- **Structural resilience** under extreme conditions, as shown by the stress test analysis, which confirms preserved diversification benefits and low beta exposure even during sharp market dislocations;
- **Model-independent stability** of results across alternative signal specifications, confirming that the performance is not driven by overfitting or specific parameter choices.

Collectively, these findings highlight the strategy’s capacity to deliver uncorrelated alpha and enhance the risk-adjusted efficiency of traditional portfolios, even with modest crypto exposures.

Limitations and Future Uncertainty

While our backtest spans a multi-year period characterized by diverse market regimes, including periods of extreme volatility and macroeconomic uncertainty, it remains inherently retrospective. Several factors may affect the persistence and viability of the proposed strategy going forward:

- **Structural evolution** in crypto markets, including increasing institutional participation and changes in liquidity profiles;
- **Regulatory developments** that may impact accessibility, tradability, or shorting of specific assets;
- **Technological shifts** (e.g., layer-2 adoption, tokenomics innovations) that could alter the behavior of valuation proxies;
- **Selection bias**, as our crypto universe was constructed ex post and may not reflect the full lifecycle of investable assets.

Despite these limitations, the methodological transparency, signal simplicity, and empirical rigor of the strategy offer a strong foundation for further research and potential real-world implementation in the evolving landscape of digital asset management.

6 Discussion

This section explains the empirical results in the context of asset pricing theory and the structural characteristics of cryptocurrency markets. Our aim is to identify the economic significance of the results and evaluate the position of the mispricing-based crypto strategy in a diversified portfolio context.

Crypto as a Factor-Driven Asset Class

The analysis discovers that cryptocurrencies, although decentralized and speculative in nature, are systematically sensitive to certain cross-sectional characteristics. Specifically, the strategy built on the mispricing factor, a blend of size and risk-adjusted momentum signals along the lines of Han et al. (2023), yields robust alpha both in standalone and portfolio-included settings. This suggests that cryptocurrency returns are not purely random or sentiment-driven, but exhibit structured return premia akin to traditional asset classes.

Mispricing factor accounts for persistent performance differences between tokens with contrasting size and momentum-adjusted profiles. This is in accordance with the bur-

geoning literature recognizing the emergence of factor structures in crypto markets, most notably as institutional participation and trading sophistication increase. Unlike conventional factors like value or quality, which are hard to define in the crypto world, size and short-horizon momentum are easily observed and economically significant even under a decentralized setup.

Efficiency or Risk Compensation?

The high and positive alpha offered by the mispricing model to CAPM, FF4, and SUR models raises a question here: are we exploiting inefficiencies or being compensated for bearing some form of risk?

The evidence suggests the former. The size and risk-adjusted momentum-based factor portfolios offer excess returns not explainable by traditional market or style exposures. The unpriced and negative beta risk in the Fama–MacBeth cross-sectional regression also lends support to this argument that the strategy is deviation-based and not compensation for systematic risk. The coin pricing models supplemented with the mispricing factor (NTFM and NTFM–SM) perform strongly compared to the base coin three-factor model, attesting to its economic relevance.

Portfolio Implications and Risk Dynamics

From a portfolio construction perspective, including the mispricing strategy within a typical 60/40 framework leads to extremely material improvements in return-to-risk tradeoffs. Sharpe and Sortino ratios increase monotonically with allocation, while drawdowns decrease (a finding contrary to the intuition that adding crypto will increase risk). This is because of the market-neutral, long–short form of the strategy and the low correlation with conventional assets.

Stress tests validate that extreme crypto crashes cause temporary tracking with the benchmark, especially at larger allocations. Yet, after shocks, betas and correlations remain at tolerable levels. This means that, while the strategy is not fully shielded from systemic shocks, its structural diversification benefits remain valid under most scenarios.

Interpreting the Mispricing Factor Mechanism

Rather than capturing behavioral biases or bubble speculation, the mispricing factor captures a quantitative arbitrage mechanism based on lagged market information and liquidity factors. The combination of size and short-horizon RA-momentum accurately identifies tokens with persistently different risk-adjusted payoff profiles. The results show that small, high-momentum coins outperform, particularly at short horizons, suggesting the presence of pricing frictions or investor inattention in less liquid segments of the market.

Moreover, the frequency of occurrence of this signal in both out-of-sample and in-sample intervals indicates that underlying return patterns are robust and are not influenced by overfitting or data mining.

Consistency with Theory

Overall, the results are in favor of a multifactor view of crypto markets in which some of the returns' predictability is due to observable characteristics. While the market remains less efficient than the traditional asset classes, it increasingly possesses structure. Mispricing factor is demonstrated to be a sound expansion of crypto asset pricing models and is of use in practice to optimize portfolios.

Rather than benefiting from irrational investor psychology, the strategy appears to be exploiting transient pricing anomalies resulting from broken liquidity, inconstant attention, and short-run overreaction: traits that will inevitably abate as the market matures.

7 Conclusions

This paper investigates the integration of a systematic, factor-based cryptocurrency strategy into traditional portfolio frameworks. Grounded in the mispricing factor developed by Han et al. (2023), a composite signal built on size and risk-adjusted momentum, we evaluate both standalone performance and the effects of inclusion in a conventional 60/40 equity–bond allocation.

Our empirical analysis yields several key findings. First, the mispricing strategy produces consistent and statistically significant alpha on a range of asset pricing models, including CAPM, FF4, and Fama–MacBeth specifications. Second, adding the strategy to hybrid portfolios enhances risk-adjusted performance, reducing drawdowns and Sharpe and Sortino ratios. These benefits are maintained across various allocation levels and are resilient to stress tests and bootstrap inference. Third, the factor has persistent predictive power over time and specifications, which highlights its structural significance rather than temporary aberration.

For investors and asset allocators, these results suggest that crypto markets, when invested in through rigorous signal-based methods, have the potential to offer diversification benefits without introducing volatility or tail risk. For policymakers, our findings help advance the emerging view of crypto assets as a factor-driven asset class, grounded in measurable economic mechanisms rather than mere speculation.

That said, several limitations demand caution. Our stress-tested and multi-year backtest is data availability- and weekly rebalancing-constrained. Shorting constraints, transaction costs, and liquidity frictions, partially dealt with in our methodology, can continue to affect implementability. Additionally, with the crypto universe’s fast-evolving nature, structural breaks and regime shifts can impact signal persistence.

Future research can be taken in a variety of directions for this framework. First, adding more signals, e.g. network effects, order book imbalance, or sentiment indicators, can make the factor model richer. Second, alternative asset pricing models involving machine learning or nonlinear dynamics can uncover latent patterns not captured by linear regressions. Third, expanding coverage to new classes of tokens (e.g., DeFi, gaming, real-world assets) can identify novel mispricing opportunities. Finally, out-of-sample verification across more tumultuous macro regimes and geographies would continue to solidify the generalizability of our findings.

Overall, the crypto strategy based on mispricing presents a compelling argument for inclusion in modern asset allocation. It demonstrates that well-structured data-driven approaches can indeed go a long way in domesticating digital assets’ volatility, serving as a bridge between traditional finance and the emerging crypto ecosystem.

A Data Collection via CryptoCompare API

All cryptocurrency data used in this study were retrieved from the CryptoCompare API (<https://min-api.cryptocompare.com>) using a registered developer key. We leveraged the `histoday` endpoint to obtain daily OHLCV (Open, High, Low, Close, Volume) data for each cryptocurrency in USD, as illustrated below:

```
https://min-api.cryptocompare.com/data/v2/histoday?fsym=BTC&tsym=USD&limit=2000
```

The data retrieval and preprocessing were fully automated in MATLAB, with robust handling of missing values, duplicates, and misaligned dates. Specific steps included:

- Forward-filling missing prices for up to 2 consecutive days to avoid data dropouts.
- Dropping assets with persistent or systematic gaps to maintain data quality.
- Filtering out stablecoins and low-liquidity assets based on volume and volatility thresholds.
- Retaining only `close` prices and `volume` (used as a proxy for market capitalization) for signal construction.
- Aggregating daily data to weekly frequency using `retime(..., 'weekly', @mean)` and `retime(..., 'weekly', @sum)` depending on the field.
- Aligning timestamps across all cryptocurrencies and benchmark assets to ensure synchronized rebalancing.

The cleaned dataset spans from October 23, 2019 to March 29, 2025, and comprises 79 cryptocurrencies that passed all quality checks. It serves as the input foundation for mispricing signal construction and strategy backtesting, as described in Sections 2.2 and 5.

All code used for this preprocessing step is available upon request and follows best practices for reproducibility and transparency in financial data analysis.

B MATLAB Code for Portfolio Construction and Econometric Analysis

All computations and empirical tests were performed in MATLAB R2024a. The codebase was developed locally and is available upon request from the authors.

The MATLAB implementation includes the following key modules:

- **Benchmark Portfolio Construction:** Simulation of the traditional 60/40 portfolio using SPY and AGG with monthly rebalancing. Daily log-returns are aggregated to weekly frequency for consistency.
- **Signal Computation:** Weekly mispricing scores are computed using two standardized signals: Size (inverse log market cap) and Risk-Adjusted Momentum (RAMOM) over 1-, 2-, and 4-week horizons. The signals are z-scored and averaged to form a composite mispricing factor.
- **Long–Short Strategy Implementation:** A market-neutral portfolio is rebalanced weekly by going long on the top 30% and short on the bottom 30% of ranked cryptocurrencies, including turnover-based transaction costs of 10 basis points.
- **Hybrid Portfolio Integration:** Portfolios are built by combining the 60/40 benchmark with increasing allocations (1% to 20%) in the crypto strategy. Performance is evaluated across cumulative returns, Sharpe and Sortino ratios, drawdowns, and rolling statistics.
- **Econometric Analysis:** Time-series regressions (CAPM and FF4), Fama–MacBeth cross-sectional models, Seemingly Unrelated Regressions (SUR), GARCH(1,1) estimation, Diebold–Mariano tests, bootstrap simulations, and structural break diagnostics (Chow, Breusch–Pagan, Durbin–Watson, Ljung–Box).
- **Stress Testing:** One-week and extended crypto shock scenarios are simulated to assess portfolio resilience and analyze post-shock correlation, beta, and drawdowns at different crypto exposure levels.

The codebase is modular and fully replicable using standard MATLAB toolboxes. Data preprocessing, signal engineering, portfolio simulation, and statistical inference are all executed within a unified environment to ensure consistency and reproducibility.

C Cryptocurrency Universe

The following 79 cryptocurrencies were included in our final dataset after filtering for data completeness, liquidity, and stability. The list includes large-cap tokens (e.g., BTC, ETH, BNB, SOL) and mid-cap assets actively traded throughout the sample window.

- | | | |
|--------|--------|---------|
| • BTC | • XMR | • SAND |
| • ETH | • ETC | • FLOW |
| • BNB | • FIL | • THETA |
| • SOL | • ATOM | • RPL |
| • XRP | • HBAR | • KAS |
| • TON | • VET | • CHZ |
| • DOGE | • ICP | • XDC |
| • ADA | • CRO | • CSPR |
| • TRX | • NEAR | • ZEC |
| • LEO | • OP | • GRT |
| • LINK | • AR | • ENS |
| • XLM | • MKR | • LDO |
| • AVAX | • XTZ | • KSM |
| • LTC | • QNT | • BAT |
| • DOT | • AAVE | • ZRX |
| • BCH | • EGLD | • ENJ |

- | | | |
|---------|--------|---------|
| • ZEN | • DCR | • LRC |
| • WAVES | • SC | • MASK |
| • QTUM | • DGB | • SUSHI |
| • ANKR | • RVN | • COTI |
| • ZIL | • TEL | • HIVE |
| • RSR | • BDX | • KDA |
| • NEO | • XEM | • GT |
| • BEAM | • ONT | • ALGO |
| • DASH | • XNO | • STX |
| • GLM | • BAND | • FET |

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